Answers and talk notes

Describe Data by Range and Interquartile Range

What your student is learning

Tonight we use range and interquartile range together to say which of two groups is more consistent.

Answers

Ask your student to explain before you look.

1. **A.** Closer together, so more consistent. Smaller spread means the values sit closer together, so the data is more consistent. So the answer is: Closer together, so more consistent.
2. **B.** Class B. 12 is smaller than 20, so Class B is more consistent.
3. **C.** Team S. 10 is greater than 6, so Team S has the more spread out middle half.
4. **D.** Range 40, IQR 12. Range = 90 − 50 = 40 and IQR = 70 − 58 = 12, so Range 40, IQR 12.
5. **A.** X has the larger range, and Y has the larger IQR. 30 > 25, so X has the larger range. 14 > 10, so Y has the larger IQR. So the answer is: X has the larger range, and Y has the larger IQR.

Talk about it

Do the range and the IQR point to the same team? What does that tell you?

Home mission

Think of two ways to get to school or the store. Which one always takes about the same time? Which one is sometimes fast and sometimes slow? Which has the smaller spread?

When they are stuck, ask

- Is this measure a subtraction? Which two numbers are we subtracting?
- Could the middle half ever be wider than the whole set?
- Which team's numbers sit closer together?

Try not to say

- Add the quartiles to find the IQR.
- A bigger range is always better.
- Range and IQR mean the same thing, just pick one.